

STAT 337: ANOVA Activity #1

Lucas Jones

1. Work through the practice in the course notes to write out the cell-means and reference-coded hypotheses, estimates, and comparison of estimates for the “CO2 Emissions for 5 Country Groups Example” (pg. 8 - 11).

Once finished with 1, complete the code chunks below and answer Questions 2-5. Once you’ve pasted and tested (i.e., run code inline to make sure it works) and answered any associated question, change `eval = FALSE` to `eval = TRUE` and knit this document.

2. Using the code chunk below, read in the `CO2_emissions_country_groups.csv` dataset. Create an appropriate data visualization, and output summary statistics (Hint: p.10 & 13 of the notes or the beginning of Section 2.2 of the textbook). Add comments describing what each line of code is doing. Make some notes about what you observe in your visualization.

```
library(tidyverse)

## — Attaching core tidyverse packages — tidyverse
## 2.0.0 —
## ✓ dplyr      1.1.4      ✓ readr      2.1.6
## ✓ forcats    1.0.1      ✓ stringr    1.6.0
## ✓ ggplot2    4.0.1      ✓ tibble     3.3.1
## ✓ lubridate  1.9.4      ✓ tidyr      1.3.2
## ✓ purrr      1.2.1
## — Conflicts —
tidyverse_conflicts() —
## ✗ dplyr::filter() masks stats::filter()
## ✗ dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
## conflicts to become errors

library(mosaic)

## Registered S3 method overwritten by 'mosaic':
##   method                        from
##   fortify.SpatialPolygonsDataFrame ggplot2
##
## The 'mosaic' package masks several functions from core packages in order
## to add
## additional features. The original behavior of these functions should not
## be affected by this.
##
## Attaching package: 'mosaic'
##
## The following object is masked from 'package:Matrix':
##
```

```

##      mean
##
## The following objects are masked from 'package:dplyr':
##
##      count, do, tally
##
## The following object is masked from 'package:purrr':
##
##      cross
##
## The following object is masked from 'package:ggplot2':
##
##      stat
##
## The following objects are masked from 'package:stats':
##
##      binom.test, cor, cor.test, cov, fivenum, IQR, median, prop.test,
##      quantile, sd, t.test, var
##
## The following objects are masked from 'package:base':
##
##      max, mean, min, prod, range, sample, sum

library(yarrr)

## Loading required package: jpeg
## Loading required package: BayesFactor
## Loading required package: coda
## *****
## Welcome to BayesFactor 0.9.12-4.7. If you have questions, please contact
## Richard Morey (richarddmorey@gmail.com).
##
## Type BFManual() to open the manual.
## *****
##
## Attaching package: 'BayesFactor'
##
## The following object is masked from 'package:mosaic':
##
##      compare
##
## Loading required package: circlize
## =====
## circlize version 0.4.17
## CRAN page: https://cran.r-project.org/package=circlize
## Github page: https://github.com/jokergoo/circlize
## Documentation: https://jokergoo.github.io/circlize\_book/book/
##
## If you use it in published research, please cite:
## Gu, Z. circlize implements and enhances circular visualization

```

```
## in R. Bioinformatics 2014.
##
## This message can be suppressed by:
## suppressPackageStartupMessages(library(circlize))
## =====
##
##
## Attaching package: 'yarrr'
##
## The following object is masked from 'package:ggplot2':
##
## diamonds

emissions5 <- read_csv("CO2_emissions_country_groups.csv")

## Rows: 120 Columns: 4
## — Column specification

```

```
## Delimiter: ","
## chr (2): Country.Name, Country.Code
## dbl (2): Year, CO2.Emissions
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this
message.

emissions5 <- emissions5 %>% drop_na()

favstats(CO2.Emissions ~ Country.Code, data = emissions5) #Computes summary
stat.

```

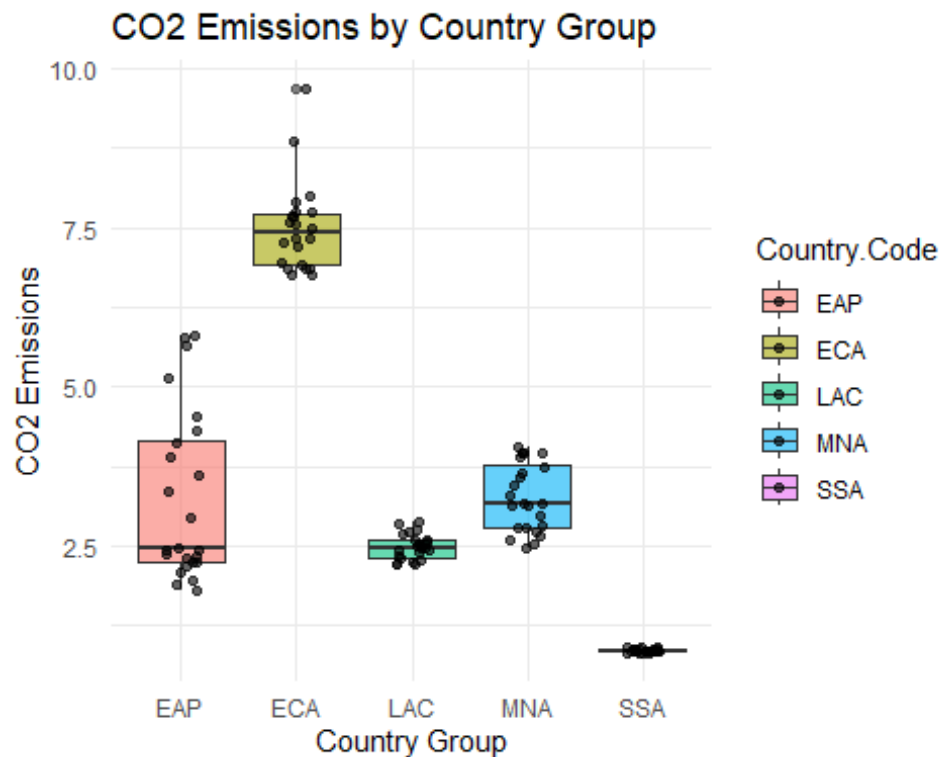
	Country.Code	min	Q1	median	Q3	max	mean
## 1	EAP	1.7979242	2.2563410	2.4511929	4.1540989	5.8028410	3.2422600
## 2	ECA	6.7439723	6.9233555	7.4080764	7.7232365	9.6769388	7.4982439
## 3	LAC	2.2027822	2.3174563	2.4727995	2.5893252	2.8896190	2.4848563
## 4	MNA	2.4602369	2.7913448	3.1699638	3.7685371	4.0657783	3.2659236
## 5	SSA	0.8106349	0.8387858	0.8517932	0.8736846	0.9183288	0.8560742

```
## sd n missing
## 1 1.33500241 24 0
## 2 0.70392745 22 0
## 3 0.19905381 24 0
## 4 0.53373390 24 0
## 5 0.02802744 24 0

ggplot(emissions5, aes(x = Country.Code, y = CO2.Emissions, fill =
Country.Code)) + #Creates Boxplot
  geom_boxplot(alpha = 0.6) + #Draws boxplot for each group
  geom_jitter(width = 0.15, alpha = 0.6) + #adds points
  labs(title = "CO2 Emissions by Country Group",
        x = "Country Group",

```

```
y = "CO2 Emissions") + # add title and axis labels
theme_minimal() #clean theme
```



- ***The dataset contains emissions grouped by country development level. Developed countries tend to have higher median CO2 emissions.*

4. Fit the reference coded model and show the summary output

```
rc_model <- lm(CO2.Emissions ~ Country.Code, data = emissions5) #creates model
summary(rc_model) #display summary
```

```
##
## Call:
## lm(formula = CO2.Emissions ~ Country.Code, data = emissions5)
##
## Residuals:
```

	Min	1Q	Median	3Q	Max
	-1.44434	-0.29118	-0.00747	0.18755	2.56058

```
##
## Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.24226	0.14734	22.005	< 2e-16 ***
Country.CodeECA	4.25598	0.21306	19.976	< 2e-16 ***
Country.CodeLAC	-0.75740	0.20838	-3.635	0.00042 ***
Country.CodeMNA	0.02366	0.20838	0.114	0.90979
Country.CodeSSA	-2.38619	0.20838	-11.451	< 2e-16 ***

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7218 on 113 degrees of freedom
## Multiple R-squared:  0.9027, Adjusted R-squared:  0.8992
## F-statistic: 261.9 on 4 and 113 DF,  p-value: < 2.2e-16
```

5. Fit the *cell-means model* and show the summary output

```
cm_model <- lm(CO2.Emissions ~ Country.Code -1, data = emissions5) # fit
cell-means model
summary(cm_model) # show summary

##
## Call:
## lm(formula = CO2.Emissions ~ Country.Code - 1, data = emissions5)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.44434 -0.29118 -0.00747  0.18755  2.56058
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## Country.CodeEAP    3.2423     0.1473   22.00 < 2e-16 ***
## Country.CodeECA    7.4982     0.1539   48.72 < 2e-16 ***
## Country.CodeLAC    2.4849     0.1473   16.86 < 2e-16 ***
## Country.CodeMNA    3.2659     0.1473   22.16 < 2e-16 ***
## Country.CodeSSA    0.8561     0.1473    5.81 5.86e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7218 on 113 degrees of freedom
## Multiple R-squared:  0.9701, Adjusted R-squared:  0.9688
## F-statistic: 733.5 on 5 and 113 DF,  p-value: < 2.2e-16
```

6. Create a term plot — choose whichever you like better from the two plots given below. Report the plot and describe in your own words what it shows.

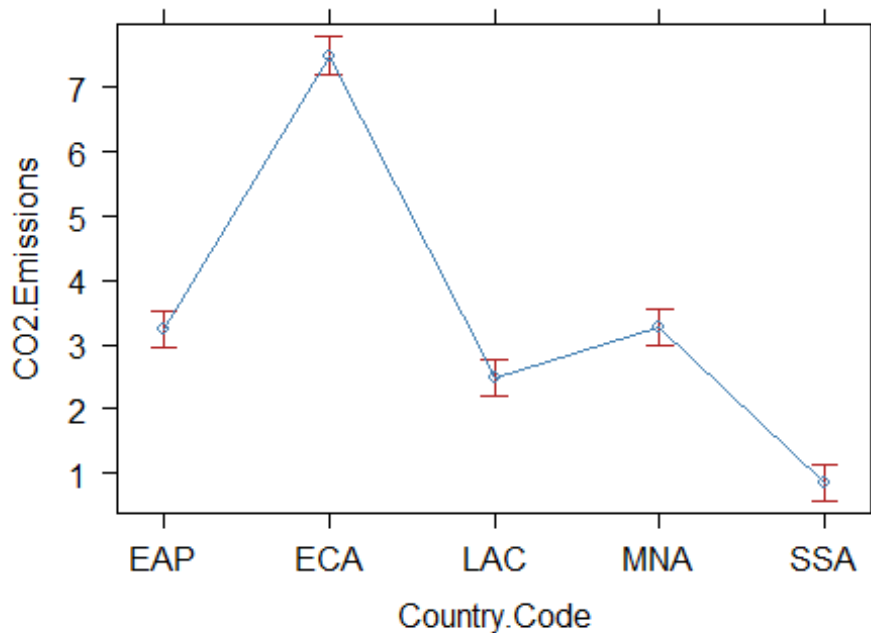
```
library(effects)

## Loading required package: carData

## Use the command
##   lattice::trellis.par.set(effectsTheme())
##   to customize lattice options for effects plots.
## See ?effectsTheme for details.

plot(allEffects(cm_model), main = "Term Plot of CO2 Emissions by Country
Group", colors=c("steelblue","firebrick"))
```

Term Plot of CO2 Emissions by Country Group



```
# you will likely need to install ggeffects to run this code
#library(ggeffects)
#ggeffect(<MODEL_FROM_Q5>, terms = "Country.Code")
#plot(ggeffect(<MODEL_FROM_Q5>, terms = "Country.Code"))
```

- The term plot shows the estimated mean CO2 emissions for each country group based on the cell-means model. Each point is the average emissions for a group, with error bars indicating the spread in the estimate. The plot clearly shows increasing Co2 emissions as country development increases.